

Yubin Kim

e: yubink.cs@gmail.com / p: 412-204-7134

HIGHLIGHTS

- 7+ yr in ML leadership, total 15+ yr in ML
- Track record of \$M business impact
- Strong ML/NLP/IR research background
- Deep search/information retrieval expert

SKILLS

Deep Learning & ML

- Transformer, LLM, VLM (e.g. BERT, LLaMA, Qwen-VL)
- Graph neural network
- Retrieval augmented generation
- LLM-based evaluation

Search/Recommendations

- A/B experimentation
- Multi-stage retrieval & ranking
- Semantic search, i.e. two-tower dense retrieval via ANN
- Query understanding
- Learning-to-rank

Tools and Frameworks

- Python, Java, C++, SQL
- Pytorch, Scikit-learn
- Lucene, Solr, Elasticsearch
- Airflow, Metaflow, Langchain
- AWS, GCP
- Cursor, Claude Code, Jupyter, vim, git

PERSONAL

- West Coast Swing dancer & teacher: *ask me about my Jack & Jill results*

Applied AI/ML leader with 7+ yrs of leadership experience building collaborative, multi-disciplinary teams to solve problems at scale in large e-commerce and healthcare organizations with a track record of significant business impact. Total 15+ yrs in machine learning, natural language processing, information retrieval, recommendations. Strong research background and active leadership roles in top tier information retrieval academic conferences and publishing venues.

EDUCATION

- Ph.D. in Computer Science from the Language Technologies Institute (Dec 2018)
Carnegie Mellon University, Pittsburgh, Pennsylvania
- Bachelor of Software Engineering (Apr 2011)
University of Waterloo, Waterloo, Ontario

PROFESSIONAL EXPERIENCE

- Chief Science Officer at **Vody, Inc.** (04/2024–current)
 - Optimized e-commerce product data feeds (e.g. Google Merchant Center) for conversational agents and better product search performance
 - Hired and led team of ~11 eng/product/ops, reported to CEO
 - Wrote and presented pitches to VCs and customers, raised 1M in funding
 - 0 → \$2M ARR, converted Grubhub, Academy Sports & Outdoors as customers
 - Owned technical and product roadmap, primary point of contact for customers
 - Utilized fine-tuned multimodal LLMs, RAG, CLIP, BERT, graph models served with vllm, ANN, kubernetes on AWS and GCP
 - Evaluated models with human-in-the-loop, LLM evaluation, and A/B testing
 - Led academic collaborations on generative engine optimization and user simulation leading to academic publications
- Senior Engineering Manager at **Etsy, Inc.** (04/2022–10/2023)
 - Built real time, large scale retrieval systems powering search, recs, and ads at etsy.com, an e-commerce marketplace with 90M+ active buyers and 100M items
 - Led product ML team (~5) of Intermediate–Sr. Staff Applied Science and Eng ICs in the Search Retrieval initiative, providing technical mentorship, managing hiring, performance, team processes
 - Directly generated \$40M+ in GMS over 1.5 years by deploying conversion-lifting innovations in graph ML models and Solr proven with A/B experiments
 - Enabled \$30M+ in ad revenue through collaborations with the Ads org
 - Co-authored the 3-year technical strategy for the Search Retrieval initiative
 - Increased the speed of online experimentation by 2x; initiated key collaborations with analytics, product, and enablement teams to develop new tooling
 - Increased rate of successful online experiments by 2x within 2 quarters, through developing robust metrics and processes for offline evaluation
- Director of Technology at **UPMC Enterprises** (07/2019–03/2022)
 - Led product team (~12) of Eng/QA ICs, Product/Eng Managers; built an Elasticsearch-based search engine over 160M+ clinical documents
 - Led applied ML team (~3) of ML engineers, responsible for creating career paths, hiring, performance management, technical mentorship

- Collaborated with product and provider/payor stakeholders to solve problems in claims processing, automated coding, risk stratification, retrospective analysis with ML: time series analysis, LSTM, CNN, decision trees
- Identified projects worth \$Ms in savings, pitched to SVPs/CTO, developed prototypes, and executed pilots with provider/payor groups
- Managed collaborations with ML/NLP healthcare start-ups and academic labs; reviewed proposals, negotiated contracts, and oversaw project delivery
- Senior Data Scientist at **UPMC Enterprises** (09/2018–07/2019)
 - Developed a search engine for social determinants of health indicators in millions of clinical documents to identify high risk patients for intervention
- Internships at Microsoft Research (2013), Amazon (2009), Google (2008)

SELECTED PUBLICATIONS

- Saber Zerhoudi, **Yubin Kim** and Michael Granitzer. 2026. [eCSM: A Complex Shopper Model for E-Commerce](#). *Under review at CIKM*.
- Yujiang Wu, Shanshan Zhong, **Yubin Kim**, Chenyan Xiong. 2026. [What Generative Search Engines Like and How to Optimize Web Content Cooperatively](#). In *Proceedings of 14th International Conference on Learning Representations (ICLR)*.
- **Yubin Kim**, Arthur Maciejewicz, Brandon Beveridge. 2025. [Lessons from the bleeding edge: large-scale production inference of LLMs](#). In *Proceedings of eCom'25: ACM SIGIR Workshop on eCommerce*.
- Jon Eskreis-Winkler, **Yubin Kim**, Andrew Stanton. 2023. [XWalk: Random Walk Based Candidate Retrieval for Product Search](#). In *Proceedings of eCom'23: ACM SIGIR Workshop on eCommerce*.
- Nithin Haridas, **Yubin Kim**. 2020. [Clustering Large-scale Diverse Electronic Medical Records to Aid Annotation for Generic Named Entity Recognition](#). In *Health Search and Data Mining Workshop*. 43–50.
- Zhuyun Dai, **Yubin Kim**, Jamie Callan. 2017. [Learning to Rank Resources](#). In *Proceedings of the 40th Annual ACM SIGIR Conference*. 837–840.
- Recent peer-reviewed workshops
 - Workshop on eCommerce, SIGIR 2023–2026
 - Workshop on Data Quality-Aware Multimodal Recommendation, RecSys 2025–2026
 - Workshop on Multimodal Search and Recommendations, ICDM 2025

Full list available on [Google Scholar](#): search "Yubin Kim" on scholar.google.com

ACADEMIC SERVICE

- Notable recent academic conference leadership
 - SIGIR 2026 Sponsorship Co-chair
 - ICTIR 2026 PC Area Chair
 - CIKM 2026 AnalytiCup Chair
 - WSDM 2026 WSDM Cup Chair
 - SIGIR 2025 Demonstration Track Co-chair
- Peer review service
 - Editorial board of Foundations and Trends in Information Retrieval (FnTIR)
 - Senior program committee of SIGIR, CIKM, WSDM, TheWebConf